AQEA

Verifiable Compute Substrate

AI Agents that cannot lie. Edge without accelerators. Search 1.85x faster.

One Substrate. Three Worlds.

Patent-pending Gen-5 Information Geometry

nextX AG - DTM Berlin - 20 May 2026

AI systems are hitting four hard walls.

Memory, energy, vendor lock-in - and now regulated AI governance - are converging at once.

Memory

Exact search at 10M+ is dominated by bytes read, not math.

Energy

Every query becomes an OpEx and emissions line item.

Lock-in

Production GPU retrieval is still largely tied to CUDA/NVIDIA.

Governance

Regulated agents need proof, not post-hoc logs.

The market wants exactness, lower cost, hardware choice - and auditability.

EU AI Act pressure adds a fourth wall: regulated agents need deterministic, replayable, verifiable behavior.

ONE SUBSTRATE - THREE WORLDS

AQEA changes the representation, not just the index.

One patent-pending substrate family powers governance, edge sensing, and hyperscale retrieval.

Constitutional Kernel

AI agents that cannot lie about what they did.

Build-time sealed governance + cryptographic audit

AQEA Shader

Hyperscale retrieval / RAG - exact and cross-vendor.

One encoded corpus. Any hardware. Same answers.

AQEA SUBSTRATE

AQEA Edge

Industrial / robotics / IoT without AI accelerators.

Sensor similarity on existing CPU / shader paths

Same substrate logic. Three wedge products. One platform narrative.

The proof: exact retrieval with a smaller footprint.

Measured on msmarco-passage / H100 class hardware; public-safe headline metrics.

1.85x
faster than Float-FAISS-GPU at 10M

23x
smaller index storage

100%
Top-K bit-identity to brute-force

1.73x
lower energy per query at 1M

4/4
GPU backends bit-identical

This is the uncomfortable part for incumbents: speed and compression are not bought by losing recall.

Why investors should care now.

AQEA is not a feature - it is infrastructure pressure relief across the AI stack.

- RAG and agent memory are growing faster than retrieval infrastructure.
- AI governance is moving from slideware to enforceable runtime evidence.
- Sovereign cloud, browser AI and edge fleets need a path that is not CUDA-only.
- Energy and index storage become board-level concerns at hyperscaler scale.

AQEA sits at the convergence: exactness + compression + determinism + hardware freedom + governance.

One encoded corpus. Any hardware. Same answers.

AQEA turns hardware choice from a lock-in decision into a deployment variable.

Strategic implication: AQEA makes hardware choice a deployment variable.

PROOF WEDGE 2 - CONSTITUTIONAL KERNEL

AI Agents that cannot lie about what they did.

Structural governance for regulated AI agents - not a wrapper, not a prompt, not policy slideware.

184/184	**14 sec**	**5**	**20**	**0**
deterministic tests passing	end-to-end deterministic demo	exhaustive outcome states	tested refusal paths	unsafe Rust in critical path

Five-Outcome Action Gate

ACCEPT_PATCH | REJECT_PATCH | REQUEST_HUMAN_REVIEW | SUPPRESS_POLICY_DENY | ESCALATE_UNCERTAIN

Build-time sealed Rule-Core + hash-chained WAL + Ed25519 approval + sandbox tree-hash. Highest precedence: SUPPRESS_POLICY_DENY.

EU AI Act ready: structural governance with a replayable cryptographic verdict trail.

PROOF WEDGE 3 - AQEA EDGE

The third wedge: edge sensing without AI accelerators.

Industrial / robotics / IoT similarity workloads can move from AI-chip paths to existing CPU / shader paths.

13/13
validated domains above floor

174%
sensor recall vs Float baseline

9.4x
best measured energy advantage

$20-80
per-device AI-chip BOM line

Industrial inspection, predictive maintenance, robot fleet health, wearables - no device audio or video leaves the edge.

The product surface is simple.

Five API surfaces turn the substrate into a deployable platform.

- **Encode** → Convert encoder output into compact deterministic artefacts.
- **Index** → Load a corpus once across cloud, browser or edge targets.
- **Query** → Return Top-K with stable ordering and reproducible results.
- **Verify** → Hash manifests and cross-platform identity for audit paths.
- **Decode** → Optional audit / general / retrieval operating modes.

Developer promise: keep your encoder and application. Replace the representation, search path and governance layer.

Four beachheads, one substrate.

Different buyers, same core pain: governed memory, lower hardware dependency, smaller indices, and deterministic outputs.

Regulated AI agents

Banks, pharma, healthcare and defense need verifiable action trails.

Hyperscale RAG

Lower index footprint and exact answers across massive corpora.

Industrial edge

Similarity workloads without dedicated AI accelerator BOM.

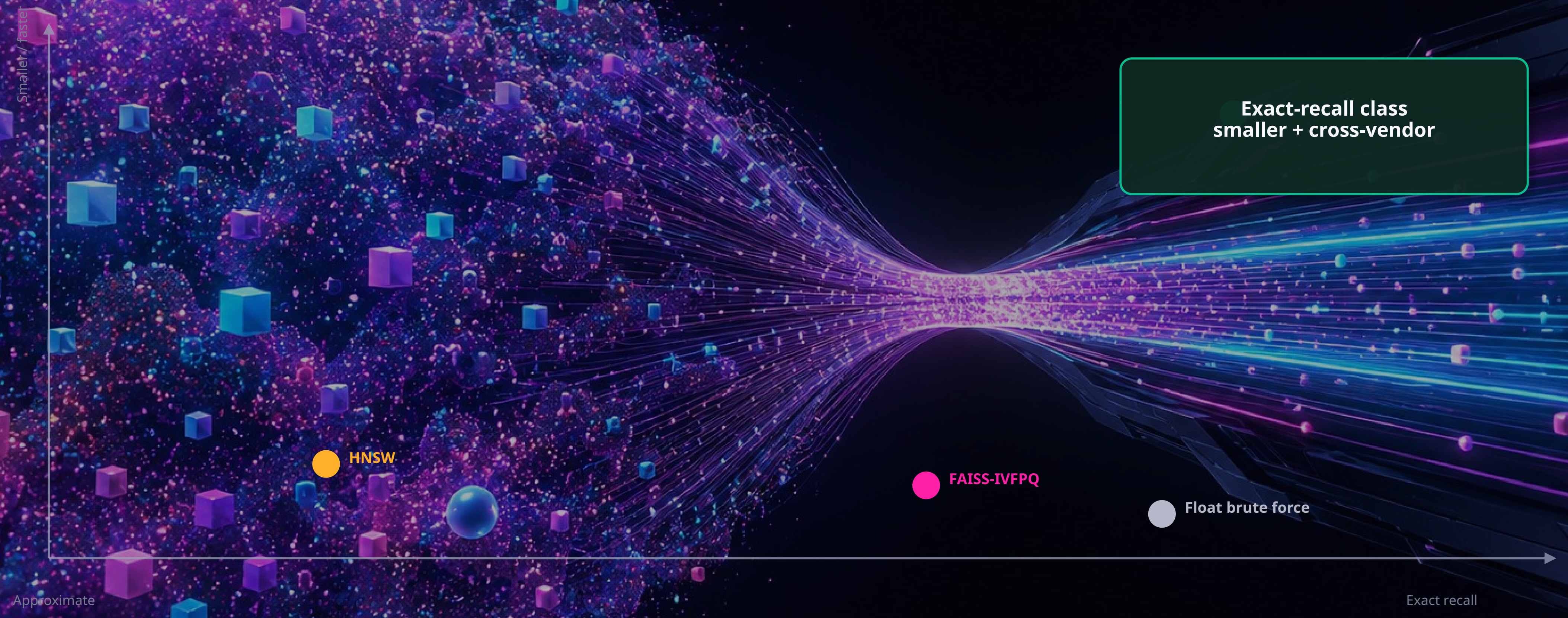
Browser / local AI

Private, local semantic search on commodity client devices.

The platform opportunity is not another vector DB. It is the representation layer underneath every vector DB.

AQEA occupies the missing quadrant.

Compression and exactness usually trade off. AQEA gives buyers a new category: exact-recall class with smaller storage.



Public-safe framing: do not expose substrate internals, quantizer details, topology, packing, or claim architecture.

Commercial motion: engineering-led, partner-first.

AQEA sells through proof, not belief. Every engagement starts with the partner workload.

1 NDA benchmark

4-6 weeks: partner corpus, partner hardware, reproducible metrics.

2 Shadow pilot

8-12 weeks: non-production integration against existing stack.

3 Strategic license

SDK, deployment rights, hardware tuning or co-development.

Revenue paths: evaluation fees -> pilot services -> annual SDK / runtime license -> co-development / strategic exclusivity.

We are looking for the partners who cannot afford to ignore this.

Capital + Strategic pilots + Hardware allies

Capital

Fund the patent runway, core engineering and GTM motion.

Strategic pilots

Run AQEA against real-world regulated, edge or RAG workloads.

Hardware allies

Validate the same artefact across GPU / CPU / browser / edge targets.

AQEA: Verifiable compute substrate for the next AI infrastructure cycle.

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